A069 v0.3.0 — Three-Dimensional Swirl Clock Emergence Exact Hill–Euler, Gross–Pitaevskii, and PKLSA-Static Controls

SST-Workbench / Development and blind provenance

18 October 2026

Abstract

We test numerical implementations of exact Euler and dimensionless nonlinear Schrodinger vortex models, along with provenance-locked static knot geometry access. Our strict target is an independently computed stable internal mode frequency for a dynamically qualified 3D knot; *that target is not reached in this release.* No measured particle clock, Lorentz ratio, Compton frequency or physical mass is provided to the solvers. The v1.0.9 framework protects blind source and output evidence.

1 Scope and epistemic status

ID A069 (provisional), v0.3.0, framework v1.0.9, minimal Python/NumPy FP64 reference. Analytic controls and E011 static metadata may pass; **physical Swirl Clock / A061 transfer remain UNRESOLVED**.

2 Question and independent physical objective

Can an internally observable, gauge-invariant, stable frequency be predicted solely from a fully 3D finite-core vortex initial state and its independently qualified dynamics? The submitted calculations do *not* yet answer this question. They test prerequisite model correctness and static source admission.

3 Competing hypotheses

H_0: no stable, identifiable 3D physical phase frequency can be predicted without a target clock fit. H_1: a precommitted mode prediction matches a held-out dynamically qualified independent provider. Both physical hypotheses remain `UNRESOLVED`; exact synthetic controls have no experimental significance.

4 Models and assumptions

Assume incompressible inviscid Euler for D1,D2A,D2H; dimensionless compressible Gross–Pitaevskii for D2N. GP and Euler are separate equations of state. Hill is an exact translating 3D Euler vortex with finite vorticity support; it is not a knotted vortex. The NLS thin-ring limit requires $R/a \gg 1$. Historical KnotPlot relaxed data are not physical inputs.

5 Symbols and dimensions

T simulation time [s], $\mathbf{u}$ velocity [m/s], $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ [1/s], Γ circulation [m^2/s], p pressure [Pa], ρ mass density [kg/m^3], a Hill sphere radius [m] or separately NLS ring core scale [m], R ring radius [m], A_H Hill vorticity gradient [1/(m s)], Ψ dimensionless GP order parameter. The nonlinear coupling and phase are dimensionless only in explicitly normalized GP units; they are not calibrated to SST constants.

6 Exact equations and independent controls

D1 and D2A (preserved from v0.2.0). D1 uses exact affine 3D Euler strain and rotating material loops; $\Gamma = 2\pi\Omega R^2$ is constant. D2A uses $u_\theta = Cr\exp(-r^2/a^2)$, Euler radial pressure $dp/dr = \rho r\Omega^2$, and independent Cartesian tracer integration. D2A has finite energy per unit axial length, not finite total three-dimensional energy.

D2H — Hill's spherical vortex. In the translating frame ($s^2 = x^2 + y^2$, $r^2 = s^2 + z^2$), for $r < a$:

$$u_x = \frac{3Uxz}{2a^2}, \qquad u_y = \frac{3Uyz}{2a^2}, \qquad u_z = \frac{3U}{2}\left(1 - \frac{2s^2 + z^2}{a^2}\right), \tag{1}$$

$$\boldsymbol{\omega} = A_H(-y, x, 0), \qquad A_H = \frac{15U}{2a^2}, \qquad \Gamma_H = \frac{2A_H a^3}{3}, \tag{2}$$

$$U = \frac{\Gamma_H}{5a} = \frac{2a^2 A_H}{15}. \tag{3}$$

Outside the sphere, the stationary co-moving flow is uniform $-U\mathbf{e}_z$ plus a decaying potential dipole:

$$(u_x, u_y) = \frac{3Ua^3 z}{2r^5}(x, y), \qquad u_z = -U + Ua^3\left(r^{-3} - \frac{3s^2}{2r^5}\right). \tag{4}$$

With $\chi = (3A_H U/4)s^2(r^2/a^2 - 1)$ inside and zero outside, $p = \rho[(U^2 - |\mathbf{u}|^2)/2 + \chi]$. Independent centered differences test $\nabla \cdot \mathbf{u} = 0$, $\nabla \times \mathbf{u} = \boldsymbol{\omega}$, and $\rho(\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p = 0$ away from the interface.

D2N — normalized GP/NLS radial core.

$$i\partial_T \Psi = -\frac{1}{2}\nabla^2 \Psi + (|\Psi|^2 - 1)\Psi, \quad \Psi = f(r)e^{i\theta}, \tag{5}$$

$$0 = -\frac{1}{2}\left(f'' + f'/r - f/r^2\right) + (f^2 - 1)f. \tag{6}$$

Boundary $f(0) = 0$ and $f(R_{\text{box}}) = 1$. A numerical boundary-value solution is checked on independent radial grids using a separately differenced differential residual. *Global $U(1)$ phase is not an operational clock.* Gauge zero-mode residual is audited, but no independently stable BdG internal mode is claimed.

D2R — thin-ring identity.

$$E = \frac{1}{2}\rho\Gamma^2 R[\ln(8R/a) - \alpha], \quad P = \rho\Gamma\pi R^2, \quad U = \frac{dE}{dP} = \frac{\Gamma}{4\pi R}[\ln(8R/a) - \beta]. \tag{7}$$

For constant a, $\beta = \alpha - 1$; for $a^2 R = \text{const}$, $\beta = \alpha - 3/2$. Roberts–Grant $\alpha \simeq 1.615$ is a reference value. Substitution $\alpha = \varphi$ is a different unverified hypothesis. The SST-69 fat-core estimate $R/a \simeq 0.898$ is *outside* the tested thin-ring regime ($R/a = 40$).

D3 — real PKLSA static admission. E011-v0.3.0 ZIP must match the pinned byte SHA-256. Its schema, PRIMARY reference status, raw and geometry hashes and geometry capability flags are inspected. A provider whose source family contains KnotPlot, legacy or VortexLab is excluded even when E011 marks it static ready. The only eligible trefoil static family is Gilbert; its full original source bytes are not embedded in the E011 ZIP, and no dynamic knot fields are supplied.

Machine-readable science contract

Research question Can full, independently qualified 3D vortex dynamics predict a gauge-invariant stable internal phase frequency from initial Euler/NLS fields alone (without Compton, Lorentz, delay input)? v0.3 tests only exact Hill Euler, radial GP and thin-ring asymptotic controls and SHA-verified E011 static eligibility.

Objective Extend reference numerics to genuinely 3D exact Hill Euler and correctly normalized GP radial vortex while adding provenance-qualified E011 static adapter. Never promote exact controls, gross phase or static topology into physical clock evidence.

H_0 No independently identified and converged 3D physical internal phase-frequency closure exists from supplied sources.

H_1 In a future preregistered DYNAMICS_READY holdout, a gauge-invariant recurring phase and blind predicted frequency converge and match across at least two truly independent physical provider families.

Registered equations

E1: Kelvin

$$\Gamma = \oint_{C(T)} \mathbf{u} \cdot d\ell = 2\pi R(T)^2 \Omega(T) \tag{8}$$

Dimensional check: m^2/s on both sides

E2: Exact affine 3D Euler

$$\mathbf{u} = A(T)\mathbf{x}, \quad A = \begin{pmatrix} -a/2 & -\Omega & 0 \\ \Omega & -a/2 & 0 \\ 0 & 0 & a \end{pmatrix}, \quad \dot{\Omega} = a\Omega \tag{9}$$

Dimensional check: 1/s times m gives m/s

E3: First-principles analytic control frequency (not knot clock)

$$\Omega(T) = \Omega_0 e^{aT} = \Gamma/[2\pi R(T)^2] \tag{10}$$

Dimensional check: 1/s

E4: Tracer phase

$$\theta(T) = \int_0^T \Omega(s)ds = \Omega_0 \,\mathrm{expm1}(aT)/a \tag{11}$$

Dimensional check: dimensionless

E5: Rotating reference frame identity

$$\boldsymbol{\omega}_{abs} = \boldsymbol{\omega}_{rel} + 2\boldsymbol{\Omega}_{frame} \tag{12}$$

Dimensional check: 1/s

E6: Euler pressure Hessian analytic control

$$\nabla\nabla p = -\rho \,\mathrm{sym}(\dot{A} + A^2) \tag{13}$$

Dimensional check: kg m^-3 s^-2

E7: Exact smooth finite-core columnar Euler flow

$$u_\theta(r) = Cre^{-(r/a)^2}, \quad \omega_z(r) = 2C(1 - r^2/a^2)e^{-(r/a)^2} \tag{14}$$

Dimensional check: [C]=1/s, [u_theta]=m/s, [omega_z]=1/s

E8: Stationary Euler pressure balance

$$\partial_r p = \rho r C^2 e^{-2r^2/a^2} \tag{15}$$

Dimensional check: Pa/m = (kg/m^3)(m)(s^-2)

E9: Independent material tracer phase check

$$\dot{\mathbf{x}} = \mathbf{u}(\mathbf{x}), \quad \theta_{\mathrm{pred}}(T) = Ce^{-(r_0/a)^2}T \tag{16}$$

Dimensional check: dimensionless phase from 1/s times s

E15: Hill exact vorticity and propagation identity

$$\omega_\phi = A_H s \ (r < a), \quad U_H = \frac{2}{15}a^2 A_H = \frac{\Gamma_H}{5a} \tag{17}$$

Dimensional check: A_H [m^-1 s^-1] * s [m] => [s^-1], a^2*A_H => [m s^-1]

E16: Stationary 2D GP m=1 vortex radial boundary problem

$$-\frac{1}{2}\left(f'' + \frac{f'}{r} - \frac{f}{r^2}\right) + (f^2 - 1)f = 0 \tag{18}$$

Dimensional check: All terms dimensionless in healing-length and inverse-chemical-potential time units.

E17: Energy impulse velocity law, with core-size derivative explicitly constrained

$$E = \frac{1}{2}\rho\Gamma^2 R(\ln(8R/a) - \alpha), \quad P = \rho\Gamma\pi R^2, \quad U = \frac{dE}{dP} \tag{19}$$

Dimensional check: E [kg m^2 s^-2]; P [kg m s^-1]; dE/dP [m s^-1].

E18: E011 static gate with immutable hash and legacy family exclusion

$$\text{accept}_{\text{static}} = \text{SHA256}_{\text{pinned}} \wedge \text{PRIMARY} \wedge \text{capabilities} \wedge \neg\text{legacy} \tag{20}$$

Dimensional check: Logical equality/selection; no physical frequency dimensions.

Registered code-to-mathematics steps

1. **S0** [G1; equations E1]: Audit provenance and refuse old relaxed KnotPlot/VortexLab data
2. **S1** [G2; equations E2, E6]: Construct divergence-free affine Euler reference and exact pressure Hessian
3. **S2** [G2; equations E1, E3, E5]: Check Kelvin loop transport and rotating reference decomposition
4. **S3** [G3; equations E2, E4]: Integrate Cartesian material tracer with RK4 and midpoint, calculate phase via atan2 unwrap
5. **S4** [G4; equations E2]: Require true 3D finite-core dynamic certification before PKLSA prediction and reveal
6. **S12** [G14; equations E2]: Load previous D1 raw JSON and compare SHA-256 with fixed signed baseline
7. **S13** [G12; equations E7, E8]: Use smooth columnar Euler exact field to calculate pressure gradient and vorticity with independent Cartesian finite differences
8. **S14** [G13; equations E7, E9]: Evolve Cartesian Lagrangian tracers with midpoint/RK4 at 160/320/640 steps and compare measured azimuth with analytic Euler prediction without introducing any clock factor
9. **S15** [G15; equations E15]: Compute exact interior/exterior Hill flow, check Euler residual and boundary continuity
10. **S16** [G16; equations E16]: Solve GP radial BVP at two radial grids, differentiate with separate stencil, audit phase-gauge null
11. **S17** [G17; equations E17]: Differentiate thin-ring E(R) numerically and compare v=dE/dP to beta laws under frozen core kinematics
12. **S18** [G18; equations E18]: SHA256 lock E011 atlas ZIP and select provider anchors by capability; reject KnotPlot and mark dynamics unresolved

7 Algorithm and software mapping

Execute D1 and D2A controls independently, then Hill exact pressure/curl diagnostics, GP radial boundary-value solution on two grids, central finite differences for dE/dP, and SHA-locked E011 parser/selector. All results serialize with explicit type (analytic, numerical GP, static geometry only). No rate predictor is fit to existing clocks. The true physics phase gate cannot be closed by a tracer angle or a global complex phase.

8 Sources and independence

S0 is inherited D1 synthetic source, byte-checked against v0.2.0. S1 is E011 v0.3.0 static atlas output metadata, pinned to archived SHA. Neither is a genuinely independent physical dynamics provider. E011 "CROSS_PROVIDER_ROBUS includes a historical KnotPlot family; this campaign disallows such legacy relaxations for physical conclusions. Provider count for potential physical source candidates therefore drops to one (Gilbert). STATIC_READY is not DYNAMICS_READY.

9 Blind gate DAG and allowed conclusions

Gates	Meaning	Allowed status
G0–G3	Protocol and inherited D1 exact controls	Analytic PASS/FAIL
G12–G14	Inherited D2A and replay checks	Analytic PASS/FAIL
G15	Exact Hill sphere field and pressure	Analytic PASS/FAIL
G16	Radial GP stationary core plus gauge-null	Model consistency PASS/FAIL
G17	Thin-ring dE/dP identities	Asymptotic PASS/FAIL
G18	E011 ZIP SHA, source-family filters	Static manifest PASS/FAIL
G19	Stable intrinsic gauge-invariant 3D eigenclock	UNRESOLVED
G4–G11	Physical 3D, PKLSA, CPU and A061 closure	NOT_APPLICABLE/UNRESOLVED

Numerical PASS does not imply physics. Invalid source hashes fail closed. Missing genuine dynamics blocks reveal; it does not falsify real physics.

10 Numerical error budget

D2H differentiates at $2\times10^{-5}a$ and requires $\|\nabla\cdot\mathbf{u}\|, \|\nabla\times\mathbf{u}-\boldsymbol{\omega}\|, \|\rho(\mathbf{u}\cdot\nabla)\mathbf{u}+\nabla p\| < 3\times10^{-8}$ at interior/exterior points, with $U-\Gamma_H/(5a) < 10^{-12}$ and $< 5\times10^{-6}$ interface approach jumps. D2N requires independent radial residual $< 2\times10^{-5}$ and between-grid profile spread $< 6\times10^{-5}$. D2R requires speed relative error $< 3\times10^{-8}$; radius ratio held at 40. These are software/model tolerances, not SST physical uncertainty bounds.

11 Backend and precision authority

NumPy FP64 + SciPy BVP are reference numerical controls only. No C++ native 3D-vortex trajectory, no SYCL certification and no Windows Arc A770 hardware qualification are claimed. DD32 remains SCREENING_ONLY. A successful Python physics surrogate cannot certify a different backend.

12 Blindness and reveal

Framework v1.0.9 generates nonced HMAC blind commitments and verifies immutable science evidence and a final output manifest. The true independent PKLSA mode holdouts do not exist as part of this artifact. Only the analyst-owned private package contains keys/nonces. No eligible physics reveal or post-hoc tuning is allowed.

13 Statistics and source strata

All analytic controls are deterministic and correlated; no p-value or significance is assigned. Three numerical radial resolutions or two integrators do not constitute two scientific providers. Historic KnotPlot cannot vote for a physical replication gate. Blind future frequency predictions need independent provider families and a frozen uncertainty metric.

14 Automatically recorded v1.0.9 results

Automatic run provenance

Field	Recorded value
Run ID	20261017T231227975712Z-e83647b2

Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	c1ea5da4b38a83650346410a85aa423b0c8495fb805ba98893fc6cd9c95ea736
Portable protocol diagnostic SHA-256	b8cfc87105bac078385bac16152d2fa8473ec84cbd888053a91a253d22e30c0d
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	af79c34d74cce83b1644ddb25a7bf64d3cef7c9f986f09a7bed3ceeaf4f536e1
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is official framework protocol integrity verified?	Framework v1.0.9 verified frozen protocol prior to science execution
G1	PASS	Were historical KnotPlot and unqualified visual relaxations excluded from scientific D1 inputs?	No historical KnotPlot or VortexLab geometry admitted as physics
G2	PASS	Does exact affine 3D Euler control satisfy divergence, Euler, Kelvin and vortex-stretching identities?	Inherited affine Euler control
G3	PASS	Does a Lagrangian Cartesian tracer yield the independently predicted control angular phase?	Inherited tracer phase control only
G12	PASS	Does the smooth finite-core columnar exact Euler control satisfy pressure, curl, incompressibility and phase tests?	Inherited columnar exact Euler only
G13	PASS	Do two time-integrators and the 160/320/640 refinement ladder reproduce the analytic columnar phase without clock fit?	Inherited time-integrator control
G14	PASS	Does the inherited D1 dataset match the preregistered original SHA-256 and independently regenerated D1 reference?	Inherited D1 exact SHA match
G15	PASS	Does exact Hill spherical 3D Euler satisfy pressure, interior curl, boundary continuity and Kelvin speed identity?	Hill sphere exact 3D Euler; not PKLSA or clock
G16	PASS	Does the dimensionless GP radial vortex solve its boundary problem with a converged residual and gauge-null check?	Dimensionless straight GP vortex, gauge phase not a physical tick
G17	PASS	Does dE/dP reproduce the thin-ring constant-core/volume velocity identities without thick-core extrapolation?	Thin-ring analytic identities only; no fat-core or physical time prediction
G18	PASS	Does E011 archive pass immutable SHA and fail-closed static provider selection without admitting KnotPlot relaxation?	E011 metadata verified; KnotPlot excluded, STATIC is NOT DYNAMICS
G19	UNRESOLVED	Do the published analytic solutions yield a qualified stable intrinsic vortex eigenmode with zero frame/gauge ambiguity?	No gauge-invariant, converged, stable intrinsic 3D vortex eigenclock
G4	NOT_APPLICABLE	Is a true finite-core 3D Euler vortex trajectory dynamically qualified?	Minimal reference profile disabled; No qualified 3D finite-core PKLSA Euler trajectory; Hill not a particle knot
G5	NOT_APPLICABLE	Is there a stable internal mode phase independent of mesh gauge and background rotation?	Minimal reference profile disabled; No stable gauge-invariant internal 3D knot eigenmode

G6	NOT_APPLICABLE	Are E010/E011 origins SHA-verified and dynamically qualified rather than only STATIC_READY?	Minimal reference profile disabled; Static atlas checked, dynamic upstream source and physical qualification absent
G7	NOT_APPLICABLE	Does an independently precommitted rate prediction match the held-out mode phase?	Minimal reference profile disabled; No physical holdout frequency precommitment
G8	UNRESOLVED	Do temporal/spatial/core and pressure-Poisson convergence, energy and helicity checks pass?	3D evolving pressure Poisson, time/space/core convergence not performed
G9	UNRESOLVED	Does provider-independent PKLSA holdout confirm signed rate and uncertainty without refitting?	Independent physics providers and phase time series absent
G10	UNRESOLVED	Has actual strict C++ FP64 certification executed for the physical numerical kernel?	Strict physical native C++ kernel certification unperformed
G11	UNRESOLVED	Is downstream A061 transfer permitted by all upstream science qualification gates?	No physical clock transfer to A061

Backend manifest


```
{
  "D1_status": "PASS",
  "D2A_status": "PASS",
  "GP_status": "PASS",
  "Hill_status": "PASS",
  "actual_backend": "python-numpy-fp64",
  "authority": "ANALYTIC_CONTROL_ONLY",
  "framework_target": "1.0.9",
  "no_native_backend_claim": true,
  "old_dataset_sha256":
"29ba1012aa3c20a3d94cf0e9898578aa0385d8f0c0986d490426035c0952f0dc",
  "physical_clock_prediction": false,
  "physical_pklsa_status": "UNRESOLVED",
  "precision": "float64",
  "schema": "SST-BACKEND-MANIFEST-2",
  "static_PKLSA_status": "PASS_STATIC_MANIFEST_ONLY",
  "thin_ring_status": "PASS"
}
```


Source manifest


```
{
  "manifest_sha256":
"f4671396fc6b7acbd3daa07d31e179ba4df22848717c6578f6529ec0534b3747",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A069",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "A069_D1_same_equations",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A069_3D_SwirlClock_Emergence\\A069-v0.3.0\\sources\\A069_v0.1.0\\D1_ANAL
YTIC_CONTROL.json",
      "provenance_family": "A069_v0.1.0_self_generated",
      "registry_hint": {
        "catalog_id": "A069",
        "hits": []
      },
      "role": "analytic_regression",
      "sha256":
"29ba1012aa3c20a3d94cf0e9898578aa0385d8f0c0986d490426035c0952f0dc",
      "source_id": "S0",
      "version": "v0.1.0"
    },
    {
      "catalog_id": "E011",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "E011_static_atlas_source_only",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
```

```
ers\\A069_3D_SwirlClock_Emergence\\A069-v0.3.0\\sources\\E011_SKLSA_Static_Re
ady_Seed_Atlas_v0.3.0-outputs.zip",
      "provenance_family": "E011_static_index_not_new_physics",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "static_geometry_diagnostic",
      "sha256":
"97cd1acbf23c102e4564f2d74fd329a2d03f32a29b8bd376ffa8b806c14246cc",
      "source_id": "S1",
      "version": "v0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Conclusions and next falsifiers

A PASS proves only that the stated numerical implementation reproduces its model's established mathematical identities. Hill supplies a genuine 3D Euler reference; GP supplies a nonlinear radial order parameter reference; neither predicts an intrinsic knot clock. For a later physical release: retrieve and verify original qualified centerlines, initialize finite-core 3D vorticity and pressure, certify long-time Euler/GP dynamics and independent material phase, then freeze an outcome-blind frequency prediction before reading held-out time series. Only then consider A061.

16 Reproduction and publication

```
python -m pytest -q tests
run_all.cmd FREEZE
run_all.cmd BASIC
run_all.cmd REPORT_ONLY
run_pklsa.cmd --topology 3_1
run_all.cmd REVEAL_IF_ALLOWED
```

This v0.3.0 is a new frozen science contract; v0.2.0 outputs are unchanged.

References

[1] M. J. M. Hill (1894), On a spherical vortex, Phil. Trans. Royal Soc. A.

[2] P. G. Saffman (1992), *Vortex Dynamics*, Cambridge University Press.

[3] P. H. Roberts and J. Grant (1971), Motions in a Bose condensate I: the structure of the large circular vortex, J. Phys. A.

[4] C. A. Jones and P. H. Roberts (1982), Motions in a Bose condensate IV: axisymmetric solitary waves, J. Phys. A.

[5] SST-Workbench, E011 SKLSA Static-Ready Seed Atlas v0.3.0 (2026), static seed contract.

[6] SST-69, SST-70, SST-72, user's research manuscripts (2026), model hypotheses not certified physics.